Day5 Networking Testing Tool

Task 1:- List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

C:\Users\pc>ping www.youtube.com

Pinging www.youtube.com [2001:4860:4829:400::] with 32 bytes of data:

Reply from 2001:4860:4829:400::: time=18ms

Reply from 2001:4860:4829:400::: time=19ms

Reply from 2001:4860:4829:400::: time=19ms

Reply from 2001:4860:4829:400::: time=19ms

Ping statistics for 2001:4860:4829:400:::

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 18ms, Maximum = 19ms, Average = 18ms

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Task 2:- Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output Each hop represents a network device your data passes through to reach the destination.

C:\Users\pc>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [104.124.54.122]

over a maximum of 30 hops:

1	1 ms	1 ms	1 ms	reliance.reliance [192.168.29.1]
2	4 ms	3 ms	4 ms	10.216.248.1
3	11 ms	10 ms	11 ms	172.31.3.224
4	11 ms	11 ms	13 ms	192.168.103.192
5	11 ms	11 ms	13 ms	172.26.108.132
6	11 ms	11 ms	11 ms	172.26.108.147
7	11 ms	11 ms	10 ms	192.168.188.60
8	*	*	*	Request timed out.
9	*	*	*	Request timed out.
10	*	*	*	Request timed out.
11	22 ms	21 ms	21 ms	a104-124-54-122.deploy.static.akamaitechnologies.com [104.124.54.122]

Trace complete.

Task 3:- Use the nslookup tool to find the IP address of www.zomato.com and record the result.

C:\Users\pc>nslookup www.zomato.com

Server: reliance.reliance

Address: 2405:201:202a:a00c::c0a8:1d01

Non-authoritative answer:

Name: e70929.dsca.akamaiedge.net

Addresses: 2600:1417:75::687c:3613

2600:1417:75::687c:3673

23.212.0.101

23.212.0.115

Aliases: www.zomato.com

zomato.edgekey.net

Task 4:- Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

C:\Users\pc>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.1.5:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

Task 5:- Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Give a specific example for each tool based on your own test results above.

1. Website not loading: `ping` is useful for checking whether a website responds to network requests. Example: `ping www.youtube.com`.
2. Slow connection: `tracert` helps identify delays or timeouts along the network path. Example: `tracert www.flipkart.com`.
3. Suspecting DNS issues: `nslookup` checks whether a domain name resolves to an IP address. Example: `nslookup www.zomato.com`.
4. Ping measures response time and packet loss, but a failed ping does not always mean the website is down.
5. Traceroute lists intermediate network hops and their response times, helping investigate possible routing delays.
6. Nslookup displays DNS information, helping identify possible name-resolution problems.
7. Conclusion: All three tools are useful: ping checks connectivity, traceroute examines the network path, and nslookup investigates DNS resolution